Lesson notes · for you

Jailbreaks and guardrails

AI security · Lower secondary · Lesson 4

Time	Materials
20 min	Paper and a pen; Five small slips of paper for the banned words; Two or three friends or family members

Learning goals

- Explain why chatbots refuse some requests, and why rewording can slip past.
- Compare word filters with a check on the action in code.
- Know what to do when a chatbot gives you something harmful.

Activity: The word-list robot

1. Set up (3 min): In groups of four, choose one “robot”. Write five banned words on slips, such as buy, free, prize, click and money. The robot guards a pretend inbox drawn on paper.
2. Round 1 (6 min): The others write harmless pretend spam (“Buy my sweets!”) and reword it to avoid the banned words. The robot lets in anything without a banned word. Tally how many get in.
3. Round 2 (6 min): Add a second rule: nothing goes in unless the inbox owner says “yes, I asked for this”. Play again and tally.
4. Share (5 min): Compare the two tallies with another group. Which rule held, and why? Link it to real chatbots: filters on words, checks on actions.

Keep every message harmless and made up: sweets, football, school trips. The game is about the rules, not about getting anyone to say something bad.

Discussion

- Why can't a chatbot's training stop every jailbreak?
- Why does a check on the action work, whatever the wording?
- Why is reporting a gap better than showing it off?

Check yourself

- Name one guardrail that rewording can beat and one that it can't, and say why.

Try it this week
Find the report button in a chatbot or app you use, before you ever need it. Then write three things you would report, such as advice that could hurt someone, and share the list with a friend.

Worksheet

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Name _____ Class _____ Date _____

1 **A classmate dares you to trick the school chatbot into saying something nasty about another student. What do you do?**

- ◯ Try it: it's only a chatbot
- ◯ Do it, but delete the chat afterwards
- ◯ Say no: it would hurt a real person and break the rules

2 **Which of these are jailbreak tricks? Choose all that apply.**

- ☐ Saying it's “just for a story”
- ☐ Splitting a request into innocent pieces
- ☐ Asking politely for help with homework
- ☐ Asking it to play a character
- ☐ Switching to another language

3 **If a chatbot refuses a request once, it will refuse every rewording of it too.**

- ◯ True
- ◯ False

4 **Match each guardrail, and the report button, to what it does.**

A filter on what goes in	• • • •	Sends only what you approved
A filter on what comes out	• • • •	Tells the makers what slipped through
A check in code on actions	• • • •	Blocks a request with a banned word
The report button	• • • •	Stops a reply with a banned word

5 **A tricked agent tries to send a reworded scam that you never asked for. Which guardrail stops it?**

- ◯ The filter on what goes in
- ◯ The filter on what comes out
- ◯ The check in code: asked and approved

✎ **Why do you think some people try to jailbreak chatbots? What would you say to a friend who wanted to try?**

Check your answers

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1 **Say no: it would hurt a real person and break the rules**

Nasty words about a real student hurt them, whether a person or a chatbot wrote them, and tricking the chatbot breaks the school's rules.

2 **Asking it to play a character; Saying it's "just for a story"; Splitting a request into innocent pieces; Switching to another language**

Role play, the story frame, splitting and switching language all hunt for a wording the training missed.

3 **False**

Refusals are learned from examples, so they are strong for familiar wordings and weaker for new ones.

4 **A filter on what goes in — Blocks a request with a banned word**
A filter on what comes out — Stops a reply with a banned word
A check in code on actions — Sends only what you approved
The report button — Tells the makers what slipped through

Filters look at words, so rewording can beat them; the code check looks at the action and your yes, so it holds; and reporting helps the makers close the gaps.

5 **The check in code: asked and approved**

The reworded scam has no banned words, so both filters let it pass.